

Signal Processing

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Disclaimer

This document contains the lectures that were supposed to be given by Dr. HOCINE, but instead he chose to teach us about electricity, anyway, hope it helps.

To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

Given the source for reading, the definitions can be somewhat improperly explained or vague as the nature of definition depends highly on context.

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Chapter 1

Fundamental Concepts

Signals are a physical representation of information that is sent from source to destination. Signal processing is the discipline of measuring, storing, and interpreting the information transmitted by signals and how to process the signals themselves in order to achieve lower noise and better transmissions.

Definition 1.1 (Signal): A signal is a real function of one or more variables, that represents a transmission of information of a physical quantity over time, space, or independent variables.

To simplify notations, in general we consider signals to be smooth.

Definition 1.2 (Noise): A noise is any unwanted interference on the signal, random or not, that affects the values of the signal when transmitted.

Notice that in the definition of noise, there is nothing specific about a certain part of signal to be noise or not, it's all up to the observer to decide.

1.1. Dirac δ Function

A really useful “function” that is often used in signal processing is the Dirac δ function. This is not really a function, it is called a distribution, which is a generalization of functions.

Definition (Support): Let $f : X \rightarrow Y$ be a function, X, Y normed vector spaces, we define the support of f as $\text{supp}(f) = \overline{\{x \in X \mid f(x) \neq 0\}}$.

Let $D(\mathbb{R}) = \mathcal{C}_c^\infty(\mathbb{R})$ the set of all smooth functions (infinitely differentiable) on a compact support. We define the distributions as follows.

Definition (Distribution): A distribution is a linear operator T from $D(\mathbb{R})$ to $\mathbb{R}$, we denote for $\varphi \in D(\mathbb{R})$, $\langle T, \varphi \rangle = T(\varphi)$.

It is quite easy to define associated a distribution for a locally integrable function for example, let f be a locally integrable function, then define the distribution T_f associated to f as $\langle T_f, \varphi \rangle = \int_{\mathbb{R}} f(x)\varphi(x) dx$. Same can be done for Radon measures, so given a measure μ , we define the distribution R_μ as follows $\langle R_\mu, \varphi \rangle = \int_{\mathbb{R}} \varphi d\mu$.

Definition 1.1.3 (Dirac δ): The Dirac δ -function is a distribution defined as follows $\delta : D(\mathbb{R}) \rightarrow \mathbb{R}$ such that $\varphi \mapsto \langle \delta, \varphi \rangle = \varphi(0)$.

We chose to define δ as a limit of a sequence of functions, and thus we write $\langle \delta, \varphi \rangle = \int_{\mathbb{R}} \varphi(x)\delta(x) dx = \varphi(0)$, and by abuse of notation we write $\forall x \neq 0, \delta(x) = 0$ and $\delta(0) = +\infty$. To generalize, define $\delta_a : \varphi \mapsto \langle \delta_a, \varphi \rangle = \varphi(a)$.

Proposition 1.1.4: Let $x(t)$ be a signal and $t_0 \in \mathbb{R}$, then

1. $\forall t \in \mathbb{R}, x(t) \cdot \delta(t - t_0) = x(t_0) \cdot \delta(t - t_0)$.
2. $\forall t \in \mathbb{R}, x(t) * \delta(t - t_0) = x(t - t_0)$.
3. $\forall t \in \mathbb{R}, \forall a \neq 0, \delta(at) = |a|^{-1} \delta(t)$.

Proof.

1. Let $t \in \mathbb{R}$. If $t \neq t_0$, $\delta(t - t_0) = 0 \Rightarrow x(t)\delta(t - t_0) = 0 = x(t_0)\delta(t - t_0)$, else, $t = t_0 \Rightarrow \delta(t - t_0) = 1 \Rightarrow x(t) = x(t_0)$ which is the since $t = t_0$.
2. We have $(x(t) * \delta(t - t_0))(s) = \int_{\mathbb{R}} x(r)\delta(s - r - t_0) dr = \int_{\mathbb{R}} x(r - t_0)\delta(s - r) dr = x(s - t_0)$, thus we get $x(t) * \delta(t - t_0) = x(t - t_0)$.
3. We have that $\int_{\mathbb{R}} \delta(at) dt = \int_{\mathbb{R}} \delta(t) \frac{1}{|a|} dt = \frac{1}{|a|} = \int_{\mathbb{R}} \frac{\delta(t)}{|a|} dt$ and hence we obtain $\delta(at) = |a|^{-1} \delta(t)$. ■

1.2. Functionals & Operations On Signals

Let $x(t), y(t)$ two signals and let $T > 0$ a window, we define the follow operations:

Scalar Product	$\langle x, y \rangle = \int_{\mathbb{R}} x(t)y(t) dt.$
Sampled Value	$x(t_0) = \langle \delta_{t_0}, x \rangle = \int_{\mathbb{R}} x(t)\delta(t - t_0) d.t$
Convolution	$(x(t) * y(t))(s) = \int_{\mathbb{R}} x(t)y(s - t) dt.$

1.3. Some Usual Functions

Sign Function	$\text{sgn}(t) = \begin{cases} 1 & \text{if } t > 0 \\ -1 & \text{if } t < 0 \\ 0 & \text{if } t = 0 \end{cases} = \begin{cases} \frac{t}{ t } & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases}$
Jump/Echelon Function	$\varepsilon(t) = \frac{1 + \text{sgn}(t)}{2} = \begin{cases} 1 & \text{if } t > 0 \\ 0 & \text{if } t < 0 \\ \frac{1}{2} & \text{if } t = 0 \end{cases}$
Ramp Function	$r(t) = \int_{-\infty}^t \varepsilon(t) dt = t \cdot \varepsilon(t)$
Rectangular Function	$\text{rect}(t) = \varepsilon\left(t + \frac{1}{2}\right) - \varepsilon\left(t - \frac{1}{2}\right) = \begin{cases} 1 & \text{if } t < \frac{1}{2} \\ 0 & \text{if } t > \frac{1}{2} \\ \frac{1}{2} & \text{if } t = \frac{1}{2} \end{cases}$
Triangular Function	$\text{tri}(t) = \text{rect}(t) * \text{rect}(t) = \begin{cases} 1 - t & \text{if } t \leq 1 \\ 0 & \text{if } t > 1 \end{cases}$
Gaussian Impulse	$\text{ig}(t) = e^{-\pi t^2}$

1.4. Quantities Of Signals

Let $x(t), y(t)$ two signals and $t_1, t_2 \in \mathbb{R}$ such that $t_1 \leq t_2$.

On Interval $[t_1, t_2]$	Average value	$\bar{x}(t_1, t_2) = \frac{1}{T} \int_{t_1}^{t_2} x(t) dt,$
	Quadratic value (Normalized energy)	$W_x(t_1, t_2) = \int_{t_1}^{t_2} x^2(t) dt$
	Average quadratic value (Normalized power)	$P_x(T) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} x^2(t) dt$
	Effective Value Of Signal	$x_{\text{eff}}(T) = \sqrt{P_x(T)}$
Globally	Running Average	$\bar{x}(t, T) = \frac{1}{T} \int_{t-T}^t x(s) ds$
	Average Value	$\bar{x} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{T/2}^{T/2} x(t) dt$
	Quadratic value (Total energy)	$W_x = \int_{-\infty}^{\infty} x^2(t) dt$
	Average quadratic value (Total power)	$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{T/2}^{T/2} x^2(t) dt$

Definition 1.4.5 (Period Sequence Of Dirac Impulses): Let $T > 0$, we define the sequence of Dirac impulses with period T as

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

Definition 1.4.6 (Repetition Operator): Let $T > 0$ and $x(t)$ a signal, the operator $\text{rep}_T(x)$ takes a function and duplicates it all over the real line $\mathbb{R}$.

$$\text{rep}_T(x)(t) = \sum_{k=-\infty}^{\infty} x(t - kT) = x(t) * \delta_T(t)$$

1.5. Signal Energy And Power

Definition 1.5.7 (Finite Energy Signal): A signal is said to have finite energy if their energy is finite, that is $W_x < \infty$.

Definition 1.5.8 (Finite Average Power Signal): A signal is said to have a finite average power if the power is finite and non-zero $0 < P_x < \infty$.

Definition 1.5.9 (Classification Of Signals): Let x be a signal, we define the following classes depending on the time and amplitude nature.

- **Analogical signal:** continuous time, continuous amplitude.
- **Quantified signal:** continuous time, discrete amplitude.
- **Sampled signal:** discrete time, continuous amplitude.
- **Numerical (Digital) signal:** discrete time, discrete time.

Definition 1.5.10 (Fourier Transform): Let $x : \mathbb{R} \rightarrow \mathbb{R}$ a function representing a signal dependent of time t . The Fourier transform of the function x is a function of frequency f , $X(f)$ such that

$$\mathcal{F}(x)(f) = X(f) = \int_{-\infty}^{\infty} x(t)e^{-i2\pi ft} dt$$

where i is the imaginary unit.

The Fourier transform is used to transform a function from the time domain to frequency/phase domain. This operation is not one way only, in many cases, we can go back to the time domain just from the frequency/phase domain, which is done by the Inverse Fourier Transform.

Definition 1.5.11 (Inverse Fourier Transform): Let $X : \mathbb{R} \rightarrow \mathbb{R}$ a function representing the strength of frequencies of a signal with parameter f . The inverse Fourier transform of the function X is a function of time $x(t)$ such that

$$\mathcal{F}^{-1}(X)(t) = x(t) = \int_{-\infty}^{\infty} X(f)e^{i2\pi ft} df$$